

10: Decision Trees

(A) $\begin{matrix} P \\ \text{(escaped)} \\ 8/14 \end{matrix} \quad \begin{matrix} P \\ \text{(not esc)} \\ 6/14 \end{matrix}$

$$G = 1 - [(8/14)^2 + (6/14)^2]$$

$$= 1 - [(0.571)^2 + (0.428)^2]$$

$$= 1 - [0.326 + 0.183]$$

$$= 1 - 0.509$$

parent
G = 0.49

(B) $\begin{matrix} \text{(urgent)} \\ \text{Left} \end{matrix}$

Right

$$G(\text{Left}) = 1 - [(5/6)^2 + (1/6)^2] = \cancel{10/36} \quad G(\text{Right}) = 1 - [(3/8)^2 + (5/8)^2]$$

$$= 1 - [(0.833)^2 + (0.166)^2]$$

$$= 1 - [0.693 + 0.027]$$

$$= 1 - 0.72$$

$$= \underline{0.28}$$

$$= 1 - [(0.375)^2 + (0.625)^2]$$

$$= 1 - [0.14 + 0.39]$$

$$= 1 - 0.53$$

$$= \underline{0.47}$$

(C) $G(\text{split}) = \frac{n_L}{n} G(\text{Left}) + \frac{n_R}{n} G(\text{Right})$

$$G = (6/14) * (10/36) + (8/14) * (30/64)$$

$$= 0.428 * 0.277 + 0.571 * 0.468$$

$$= 0.119 + 0.268$$

$$= \underline{0.387}$$

continuation on next page

Gini impurity continuation

$$\text{Gain} = G(\text{parent}) - G(\text{split})$$

$$= 0.489 - 0.387$$

$$= 0.102$$

- ① gain 0.08 of the new question is lower than the original questions' gain of 0.103 ($0.103 > 0.08$) So the tree will ask the same original question; "does the ticket contain the word 'urgent'?"

Question 1: $\frac{1}{5} = \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5}$

~~1. AB Post-Bankruptcy~~

~~Handwritten scribbles~~

$$MSE(Q) = \frac{\sum (y - \bar{y}_Q)^2}{n}$$

Question 15

(A) $\bar{y} = (5+7+15+17+19+21)/6$
 $= \frac{84}{6}$
 $= 14$

SSE $(5-14)^2 = (-9)^2 = 81$
 $(7-14)^2 = (-7)^2 = 49$
 $(15-14)^2 = 1^2 = 1$
 $(17-14)^2 = 3^2 = 9$
 $(19-14)^2 = 5^2 = 25$
 $(21-14)^2 = 7^2 = 49$

$SSE = 214$
 $MSE = \frac{214}{6} = 35.667$

(B)

~~scribbled out~~

$\bar{y}_{left} = \frac{5+7}{2} = 6$

$\bar{y}_{right} = \frac{15+17+19+21}{4}$
 $= \frac{72}{4} = 18$

$SSE_{left} = (5-6)^2 + (7-6)^2$
 $= (-1)^2 + 1^2$
 $= 2$

$MSE_{left} = \frac{2}{2} = 1$

$SSE_{right} = (15-18)^2 + (17-18)^2$
 $+ (19-18)^2 + (21-18)^2$
 $= (-3)^2 + (-1)^2 + 1^2 + 3^2$
 $= 9 + 1 + 1 + 9$
 $= 20$

$MSE_{right} = \frac{20}{4} = 5$

... continuation of 15 B

$$\text{Weighted MSE} = \frac{2}{6} * 1 + \frac{4}{6} * 5$$
$$= 0.333 + 3.333$$
$$= \underline{\underline{3.667}}$$

Gain:

$$35.667 - 3.667 =$$
$$= \underline{\underline{32}}$$

(C)

yes calls

no calls

7
17
19
21

5
15

$$\bar{y}_{\text{yes}} = \frac{7 + 17 + 19 + 21}{4}$$
$$= \frac{64}{4}$$
$$= \underline{\underline{16}}$$

$$\bar{y}_{\text{no}} = \frac{5 + 15}{2}$$
$$= \underline{\underline{10}}$$

$$SSE_{\text{yes}} = (7 - 16)^2 + (17 - 16)^2 + (19 - 16)^2 + (21 - 16)^2$$
$$= (-9)^2 + 1^2 + 3^2 + 5^2$$
$$= 81 + 1 + 9 + 25$$
$$= \underline{\underline{116}}$$

$$SSE_{\text{no}} = (5 - 10)^2 + (15 - 10)^2$$
$$= (-5)^2 + 5^2$$
$$= \underline{\underline{50}}$$

$$MSE_{\text{yes}} = \frac{116}{4} = \underline{\underline{29}}$$

Weighted

$$MSE = (4/6) * (29) + (2/6) * (25)$$
$$= 19.333 + 8.333$$
$$= \underline{\underline{27.667}}$$

$$\text{Gain} = 35.667 - 27.667$$
$$= \underline{\underline{8}}$$

$$MSE_{\text{no}} = \frac{50}{2} = \underline{\underline{25}}$$

(D)

Gain A > Gain B, so tree splits at issues ≤ 3.

each of the 2 leaves predict the mean

of its calls

question 16:

Variance:

$$\begin{aligned} X_1 \text{ Speed: } 5, 15, 15, 25 &\rightarrow \bar{X}_1 = \frac{60}{4} = 15 \\ (5-15)^2 + (15-15)^2 + (15-15)^2 + (25-15)^2 \\ &= (-10)^2 + 10^2 \\ &= 200 \rightarrow \frac{200}{4} = \underline{\underline{50}} \end{aligned}$$

$$\begin{aligned} X_2 \text{ Stops: } 10, 30, 50, 30 \\ \bar{X}_2 = \frac{120}{4} = 30 \\ (10-30)^2 + (30-30)^2 + (50-30)^2 + (30-30)^2 \\ &= (-20)^2 + 20^2 \\ &= 800 \rightarrow \frac{800}{4} = \underline{\underline{200}} \end{aligned}$$

X_3 Idle: 5, 15, 15, 5

$$\begin{aligned} \bar{X}_3 = \frac{40}{4} = 10 \\ (5-10)^2 + (15-10)^2 + (15-10)^2 + (5-10)^2 \\ &= (-5)^2 + 5^2 + 5^2 + (-5)^2 \\ &= 100 \rightarrow \frac{100}{4} = \underline{\underline{25}} \end{aligned}$$

(B) Total Variance = $50 + 200 + 25 = 275$

(C) Driver 1: $0.6 \times 5 + 0.8 \times 10 + 0 \times 5 = 3 + 8 = \underline{\underline{11}}$
Driver 2: $0.6 \times 15 + 0.8 \times 30 + 0 \times 15 = 9 + 24 = \underline{\underline{33}}$
Driver 3: $0.6 \times 15 + 0.8 \times 50 + 0 \times 15 = 9 + 40 = \underline{\underline{49}}$
Driver 4: $0.6 \times 25 + 0.8 \times 30 + 0 \times 5 = 15 + 24 = \underline{\underline{39}}$

$\frac{(11+33+49+39)}{4} = \frac{132}{4} = 33$

$$\begin{aligned} (11-33)^2 + (33-33)^2 + (49-33)^2 + (39-33)^2 \\ &= (-22)^2 + 0^2 + 16^2 + 6^2 \\ &= 484 + 256 + 36 \\ &= 776 \end{aligned}$$

$$\begin{aligned} \text{Var}(PC_4) &= \frac{776}{4} \\ &= 194 \end{aligned}$$

$$\text{Share} = \frac{194}{275} = 0.705$$

Var explaining 70.5% of

~~total variance~~ total variance